

Princeton Liu

510-335-9673 | hiprincetonliu@gmail.com | princetonliu.github.io | linkedin.com/in/princeton-liu | github.com/princetonliu

EDUCATION

University of California, Los Angeles

August 2023 – June 2027

B.S. Computer Science and B.S. Applied Mathematics, GPA: 3.9/4.0 (continuing to M.S. CS, Fall 2027)

- **Graduate Coursework:** NLP, Theoretical CS, AI for SciComp, Quantum Algorithms, Dynamic Programming
- **Coursework:** Machine Learning, Algorithms, Data Structures, Compilers, Probability, Statistics

EXPERIENCE

UCLA Department of Mathematics

June 2026 – Present

Undergraduate Researcher (Computational and Applied Math REU)

[Project Slides](#)

- Proposed a lifted Multi-FNO to predict fluid flows across viscosities **~4,600x faster** than numerical solvers
- Cut L2 error from **16.8%** to **9.0%** and wrong-rotation rate from 1.7% to 0.27% by fusing branches in latent space
- Diagnosed branch collapse in 4-branch Multi-FNOs and capped branch weights to force mixing (16.8% to 15.9%)
- Reduced 5K-trajectory dataset generation time by **3.2x** using an 8-worker pseudo-spectral pool on HDF5

UCLA Computational Vision and Learning Lab

September 2024 – September 2025

Undergraduate Researcher

[Project Slides](#)

- Injected mean **CLS activations** harvested from a fine-tuned CLIP into a frozen copy across all 12 encoder layers
- Raised frozen CLIP ViT-B/16 **same-different accuracy** from **0.49** to **0.65** without updating any parameters
- Located relational-reasoning circuits by ranking the **average indirect effect** (AIE) across all 144 attention heads

PROJECTS

MiniJava to RISC-V Compiler | *Java 8, JavaCC/JTB, TeaVM*

[Demo](#) | [Github](#)

- Built **5-stage** MiniJava to RISC-V compiler **pipeline**: parser, type checker, IR, register allocator, code generator
- Deployed **in-browser demo** of the MiniJava to Sparrow IR front end by compiling the compiler to JS via TeaVM

Inference & Visualization System for CLIP | *PyTorch, Golang, Docker, AWS EC2, FastAPI*

[Github](#)

- Exposed all **144 attention heads** via weight-recording **modules**, rendering CLS scores as 14×14 heatmaps
- Built **Go gateway** routing uploads to a FastAPI CLIP worker, shipped as Docker Compose stack on EC2

Context-Aware LLM Content Moderation | *Python, Llama 3, HateBERT*

[Report](#) | [Github](#)

- Built **dual-model system** pairing a chatbot with an **LLM supervisor** screening rolling conversation history
- Raised early escalation detection from **24%** to **81%** over the top-F1 baseline (HateBERT) using an 8B supervisor

Tokenization Schemes for Autoregressive PDE Transformers | *PyTorch, Transformers*

[Report](#)

- Compared five tokenizers for autoregressive Lorenz prediction and found a **19x** gap in per-step loss across models
- Showed the top per-token-accuracy scheme (0.903) drifted off the Lorenz attractor with **10x worse** rollout error

Neural Speech Decoder | *PyTorch, GRUs, Attention, CTC*

[Report](#)

- Lowered phoneme error rate from **0.223** to **0.167** with residual attention, time masking, and CTC smoothing
- Trained **GRU-CTC** decoders on 10,880 sentence trials, 256 neural features per 20 ms bin from 128 electrodes

Exact Dynamic Programming vs. Deep RL on Finite-Horizon MDPs | *Python, PyTorch, RL*

[Report](#)

- Modeled optimal execution as a **finite-horizon MDP** over 6,331 episodes, solved exactly by backward induction
- Trained a **Double DQN** with experience replay, target networks, and reward shaping on feature-engineered states

TECHNICAL SKILLS

Languages: Python, C++20, Java 8, Go, C, SQL, JavaScript, MATLAB, Haskell

ML & Research: PyTorch, Scikit-learn, NumPy, Pandas, deep learning, computer vision, NLP, reinforcement learning

Backend & Infrastructure: REST APIs, FastAPI, Go net/http, Docker, Nginx, AWS EC2, Git, Linux

AWARDS AND TEACHING

Awards: 2022 USAMO Qualifier (**top 250** out of 50,000 AMC 12 contestants)

Undergraduate Grader: Machine Learning, Programming Languages, Computer Organization, Logic Design

Undergraduate Teaching Assistant: Data Science, Probability